

BRSM Movie Memory Experiment

Inferential Statistical Report

Team RSS

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1 Sample and Design

Between-subjects design. After the updated cleaning pass: **170 participants** (Natural Boundary [NB], $n = 89$; Abrupt Boundary [AB], $n = 81$), 6,800 recognition trials total (40 per participant; 20 Before-Boundary [BB] and 20 Event-Middle [EM]). Each participant saw 40 short videos; the recognition task presented one target–lure forced-choice pair per video. Every movie contributes exactly one trial per participant.

2 Analytical Framework

2.1 Recognition Memory Index (REC)

All accuracy-based DVs are expressed on the **Recognition Memory Index (REC)**. In each recognition trial a single previously-seen target frame is paired with a single unseen lure frame; the participant indicates which is the previously-seen frame. Define:

$$H = P(\text{"target"} \mid \text{target present}) = \text{accuracy} \quad FA = P(\text{"target"} \mid \text{lure present}) = 1 - \text{accuracy}$$

where the second equality holds because each forced-choice trial contains exactly one target and one lure, so the two probabilities sum to one. The experiment brief’s formula $REC = P(\text{"old"} \mid \text{Target}) - P(\text{"old"} \mid \text{Foil})$ therefore becomes

$$\boxed{REC_i = H - FA = \text{accuracy}_i - (1 - \text{accuracy}_i) = 2 \cdot \text{accuracy}_i - 1} \quad \in [-1, +1], \quad 0 = \text{chance}.$$

This is formally identical to Snodgrass & Corwin’s (1988) corrected recognition (P_r). Because REC is a monotone affine transform of accuracy, all rank-based test statistics (Mann–Whitney U , p , r_{rb}) and Shapiro–Wilk verdicts are *identical* to those on raw accuracy; only the descriptive scale changes.

2.2 Log-Transformation of Response Time

Trial-level raw RT is strongly right-skewed (skew = 3.83, excess kurtosis = 31.6). All RTs are log-transformed ($RT^* = \ln(RT)$) before analysis; verification is reported in Section 5.1.

2.3 Test-Selection Decision Rule

1. **Shapiro–Wilk** ($\alpha = .05$) within each group. If either group departs from normality: **Mann–Whitney U test** (two-sided, continuity-corrected); effect size = rank-biserial $r_{rb} = 2U_1/(n_1n_2) - 1$ (positive values support the directional prediction).
2. If both groups are normal: **Levene’s test** (median-centred, Brown–Forsythe). Equal variances ($p \geq .05$): **Student’s t -test**; unequal: **Welch’s t -test**. Effect size = Cohen’s d with Hedges’ g correction.

All five primary tests are corrected with the **Benjamini–Hochberg (BH) false-discovery-rate procedure** ($m = 5$, $\alpha_{FDR} = .05$). BH is applied to two-sided p -values; one-sided p -values are reported for directional hypotheses throughout.

3 Primary Hypotheses

3.1 H1 — Overall Recognition Memory (NB > AB)

Hypothesis. Natural Boundary participants show higher overall recognition memory than Abrupt Boundary participants.

Group	n	M	SD	Mdn	$[Q_1, Q_3]$
NB	89	0.743	0.144	0.750	[0.650, 0.850]
AB	81	0.679	0.164	0.700	[0.550, 0.800]

Table 1: H1 descriptives (REC scale).

DV. Per-participant $REC = 2 \cdot (\text{correct}/40) - 1$.

Normality (Shapiro–Wilk). NB: $W = 0.940$, $p = 4.5 \times 10^{-4}$; AB: $W = 0.954$, $p = 5.7 \times 10^{-3}$. Both non-normal (ceiling-bounded, left-skewed) $\Rightarrow$ **Mann–Whitney U test.**

Result. $U = 4438.5$, $p_{\text{one-sided}} = .0044$, $p_{\text{BH}} = .022$, $r_{rb} = +0.231$.

Decision. Reject H_0 . NB > AB; small-to-medium effect; survives BH correction.

3.2 H2 — Before-Boundary REC (NB > AB)

Hypothesis. Recognition of target frames appearing immediately before a consensus event boundary is higher in the NB condition.

DV. $REC_{\text{BB}} = 2 \cdot (\text{correct BB}/20) - 1$.

Group	n	M	SD	Mdn
NB	89	0.719	0.192	0.800
AB	81	0.648	0.209	0.700

Table 2: H2 descriptives (REC_{BB} scale).

Normality (Shapiro–Wilk). NB: $W = 0.919$, $p = 3.4 \times 10^{-5}$; AB: $W = 0.946$, $p = 1.9 \times 10^{-3}$. Both non-normal $\Rightarrow$ **Mann–Whitney U test.**

Result. $U = 4375.0$, $p_{\text{one-sided}} = .0075$, $p_{\text{BH}} = .025$, $r_{rb} = +0.214$.

Decision. Reject H_0 . NB BB-frame recognition significantly higher; consistent with Swallow, Zacks & Abrams (2009). Survives BH correction.

3.3 H3 — Mean Confidence Rating (NB > AB)

Hypothesis. NB participants report higher mean confidence in their recognition responses.

DV. Per-participant mean Likert confidence rating (1–5 scale).

Group	n	M	SD	Mdn
NB	89	4.201	0.414	4.275
AB	81	4.081	0.457	4.100

Table 3: H3 descriptives (confidence rating, 1–5).

Normality (Shapiro–Wilk). NB: $W = 0.971$, $p = .045$; AB: $W = 0.975$, $p = .116$. NB fails ($p < .05$; ceiling-compressed distribution) $\Rightarrow$ **Mann–Whitney U test.**

Result. $U = 4150.5$, $p_{\text{one-sided}} = .044$, $p_{\text{BH}} = .111$, $r_{rb} = +0.152$.

Decision. Fail to reject H_0 . Direction matches prediction but does not survive BH correction. Strong ceiling effect (most ratings 4–5) limits sensitivity.

3.4 H4 — Proportion of Low-Confidence Trials (AB > NB)

Hypothesis. AB participants produce a higher proportion of low-confidence trials (confidence rating ≤ 3).

DV. Per-participant proportion of trials with confidence ≤ 3 .

Group	n	M	SD	Mdn	$[Q_1, Q_3]$
AB	81	0.265	0.147	0.275	[0.150, 0.375]
NB	89	0.235	0.127	0.225	[0.150, 0.300]

Table 4: H4 descriptives (proportion low-confidence, $\text{conf} \leq 3$).

Normality (Shapiro–Wilk). AB: $W = 0.976$, $p = .137$; NB: $W = 0.973$, $p = .062$. Both groups pass.

Equal variances (Levene, median-centred). $F(1, 168) = 3.11$, $p = .080$ — equal variances assumed $\Rightarrow$ **Student’s independent-samples t -test.**

Result. $t(168) = 1.452$, $p_{\text{one-sided}} = .074$, $p_{\text{BH}} = .148$, Cohen’s $d = +0.223$, Hedges’ $g = +0.222$.

Decision. Fail to reject H_0 . Both groups pass normality with the ≤ 3 cut-off, enabling a parametric test, but the between-group difference remains small and non-significant.

3.5 H5 — Per-Movie Accuracy Variability (AB > NB)

Hypothesis. AB participants show greater trial-to-trial variability in recognition memory across movies.

DV. Per-participant standard deviation ($\text{ddof} = 1$) of per-movie accuracy across the 40 movies. Because each (participant, movie) cell contains exactly one Bernoulli outcome, per-movie accuracy $\in \{0, 1\}$ and this SD is a direct measure of within-participant inconsistency.

Group	n	M	SD	Mdn
AB	81	0.354	0.080	0.362
NB	89	0.320	0.086	0.335

Table 5: H5 descriptives (SD of per-movie accuracy).

Normality (Shapiro–Wilk). AB: $W = 0.954$, $p = 5.5 \times 10^{-3}$; NB: $W = 0.954$, $p = 3.1 \times 10^{-3}$. Both non-normal $\Rightarrow$ **Mann–Whitney U test.**

Result. $U = 4478.0$, $p_{\text{one-sided}} = .0032$, $p_{\text{BH}} = .022$, $r_{rb} = +0.242$.

Decision. Reject H_0 . AB participants show significantly greater per-movie accuracy variability; survives BH correction. Note: because per-movie accuracy is Bernoulli, this DV is structurally related to H1 (overall accuracy) and should not be treated as fully independent evidence.

4 Multiple-Comparison Correction Summary

BH procedure applied across five primary two-sided p -values ($m = 5$, $\alpha_{\text{FDR}} = .05$).

Three of five primary hypotheses (H1, H2, H5) survive BH correction.

	Hypothesis (direction)	Test	Statistic	$p_{\text{one-sided}}$	p_{BH}	Effect size	Survives $\alpha=.05$?
H1	Overall REC, NB > AB	Mann–Whitney U	$U = 4438.5$	.0044	.022	$r_{rb} = +0.231$	✓
H2	BB-frame REC, NB > AB	Mann–Whitney U	$U = 4375.0$	.0075	.025	$r_{rb} = +0.214$	✓
H3	Mean confidence, NB > AB	Mann–Whitney U	$U = 4150.5$	.044	.111	$r_{rb} = +0.152$	×
H4	Low-conf (≤ 3) prop., AB > NB	Student’s t	$t(168) = 1.452$	.074	.148	$d = +0.223$	×
H5	Per-movie acc. SD, AB > NB	Mann–Whitney U	$U = 4478.0$	.0032	.022	$r_{rb} = +0.242$	✓

Table 6: Primary hypothesis summary. p_{BH} is the BH-adjusted two-sided p -value (BH is applied to two-sided ps by convention). ✓ = survives correction at $\alpha = .05$; × = does not.

5 Secondary Analysis — Response Time

5.1 Log-Transform Verification

Shapiro–Wilk at the participant level:

Group	Raw W	Raw p	lnRT W	lnRT p
NB ($n = 89$)	0.901	5.1×10^{-6}	0.988	.605
AB ($n = 81$)	0.898	8.8×10^{-6}	0.972	.068

Table 7: Shapiro–Wilk: raw vs. log-transformed participant-level mean RT. Both groups pass on lnRT; both reject on raw RT.

Both groups pass on lnRT. Levene’s test: $F(1, 168) = 0.237$, $p = .627$ (equal variances) $\Rightarrow$ **Student’s t -test on lnRT**. Diagnostic plots saved to `rt_normality_participant_level.png` and `rt_normality_trial_level.png`.

5.2 Participant-Level Log-RT Test

NB: $M_{\text{lnRT}} = 1.609$, $SD = 0.237$ (geometric mean ≈ 5.00 s); AB: $M_{\text{lnRT}} = 1.573$, $SD = 0.223$ (geometric mean ≈ 4.82 s).

$t(168) = 1.013$, $p = .312$ (two-sided), Cohen’s $d = +0.156$, Hedges’ $g = +0.155$. The null is not rejected.

6 Conclusions

H1, H2, and H5 survive BH correction. Natural Boundary participants have significantly higher overall REC ($r_{rb} = +0.231$), higher before-boundary REC ($r_{rb} = +0.214$), and Abrupt Boundary participants show greater per-movie accuracy variability ($r_{rb} = +0.242$), all at $p_{\text{BH}} \leq .025$.

H3 and H4 (confidence) are not significant after correction. Both trend in the predicted direction ($r_{rb} \approx 0.15$, $d \approx 0.22$), but ceiling compression of the Likert ratings (most responses are 4 or 5) limits sensitivity.

The RT test does not reach significance on lnRT at the participant level ($p = .312$). The log-transform is verified to rescue normality (Shapiro p moves from $\sim 10^{-6}$ on raw RT to .61/.07 on lnRT).